Auteur: Ibrahim Akiyev
Studierichting: Informatica
Oefeningenreeks 2E nr. 10.20

$$2^{2x-1} - 5 \cdot 2^{x-1} + 3 = 0 \Rightarrow n = 2^{x-1}$$

$$\Leftrightarrow 2^{2x-1} = \frac{2^{2x}}{2} \Rightarrow 2^{2x-1} = \frac{4n^2}{2} = 2n^2$$

$$\Leftrightarrow 2n^2 - 5n + 3 = 0$$

$$\Leftrightarrow D = (-5)^2 - 4 \cdot 2 \cdot 3 = 1$$

$$\Leftrightarrow n = \frac{5 \pm 1}{4} \Rightarrow n = \frac{3}{2} \vee n = 1$$

$$\Leftrightarrow x = 1 \vee x = \log_2 3$$

References